



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA
Faculty of Engineering, Built Environment and
Information Technology

School of Information Technology

Department of Computer Science

COS326 – Database Systems

Practical 1 2026

Release Date: 31 July 2026
Submission Date and Time: 10 August 2026 @ 23:59Hrs
Lecturer: Mr S.M Makura
Total: 50 Marks

A. Objectives

Demonstrate that you can use ObjectDB for Java to insert, modify, retrieve and delete objects from an ObjectDB database and that you understand ObjectDB.

In this practical you will have to write a Java program that:

1. Implements Java classes and makes them persistent by storing them in an ObjectDB database.
2. Models relationships between persistent objects.
3. Executes queries on the ObjectDB database.

B. Submission Procedure

1. Create a folder as follows:

The folder name should be uXXXXXXXXX_Surname_Initials(XXXXXXXXXX is your student number) *e.g u12345678_Smith_JN* and should contain the project files (your.jar, .java and other necessary files needed for your project to run). Create a zip file of the folder and upload it to ClickUP via the submission link provided for Practical 1.

NO LATE submissions will be accepted after the submission date and time has lapsed. Do not wait until the last minute to submit and then provide reasons related to technical difficulties.

Academic Integrity: You may use AI-assisted tools (e.g., ChatGPT, Copilot, Gemini etc) to help you understand ObjectDB concepts and Java syntax. However, you remain responsible for the design, implementation, and correctness of your solution. During marking, you may be asked to explain or modify any part of your submission. Failure to demonstrate understanding may result in marks being deducted.

Question 1: ObjectDB Database Programming

Scenario

The Faculty of Engineering and the Built Environment at the University of Pretoria manages a large collection of specialised laboratory equipment used by postgraduate students and researchers. The current booking process is managed manually and has resulted in duplicate bookings, missing records and difficulties tracking equipment usage.

You have recently been appointed as a software engineer and have been tasked with developing a desktop application that will manage research equipment bookings using an ObjectDB database. The application must allow users to manage researchers, equipment and equipment bookings through a Java Swing or JavaFX graphical user interface.

Source: Makura S.M (2026)

Create a Java application using NetBeans or any Java IDE you are comfortable with. The application must interact with an ObjectDB database and must satisfy the following requirements.

(a) Classes

Your application must include classes that store information relating to:

Researcher

At minimum, the class must store:

- Researcher ID
- Full Name
- Department
- Email Address

Equipment

At minimum, the class must store:

- Equipment ID
- Equipment Name
- Equipment Category
- Purchase Date
- Replacement Cost
- Equipment Status

Booking

At minimum, the class must store:

- Booking ID
- Booking Date
- Start Time
- End Time
- Research Purpose

- Booking Status

(b) Relationships

The system must model the relationships between the classes appropriately using JPA relationship annotations.

The relationships selected must accurately represent the problem domain.

Incorrect relationship modelling will result in a loss of marks.

(c) Functional Requirements

Your application must provide a Java Swing or JavaFX GUI that enables users to perform the following operations.

(i) Register Researcher

Capture and store researcher information.

(ii) Register Equipment

Capture and store equipment information.

(iii) Create Booking

Create and store equipment bookings.

Bookings may only reference existing records.

(iv) Search Researcher

Search for a researcher using the Researcher ID.

Display the researcher together with all associated booking information.

(v) Search Equipment

Search for equipment using the Equipment ID.

Display the equipment together with its booking history.

(vi) Update Booking

Allow users to modify the status of an existing booking.

(vii) Cancel Booking

Remove an existing booking from the database.

Cancelling a booking must not remove any associated researcher or equipment records.

(viii) Equipment Summary

Display the total replacement cost of all equipment currently stored in the database.

(d) Additional Functionality

Implement the following functionality.

1. Display all equipment currently available for booking.
2. Display all bookings associated with a specified researcher.
3. Display all equipment that has never been booked.
4. Display the researcher(s) with the highest number of bookings.

(e) Business Rules

Your application must enforce the following business rules.

1. Equipment marked as **Out of Service** may not be booked.
2. A booking must have a valid start and end time.
3. A researcher may not have more than **three Active bookings** at any given time.
4. Duplicate Researcher IDs are not permitted.
5. Duplicate Equipment IDs are not permitted.
6. Duplicate Booking IDs are not permitted.
7. Conflicting bookings for the same equipment are not permitted.
8. A researcher may not create multiple bookings for the same equipment on the same date.
9. Bookings may only be created for the current date or a future date.

(f) General Requirements

Your application must:

- Perform appropriate input validation.
- Handle exceptions appropriately.
- Display meaningful error messages using Java Swing or JavaFX components.
- Demonstrate good object-oriented programming practices.

Applications that terminate due to unhandled exceptions or fail to satisfy the requirements of this specification will lose marks.

Students are expected to design an appropriate graphical user interface and determine the necessary program structure to satisfy the requirements of this specification.

Full marks will be awarded for a fully functional Java application that satisfies all the requirements of this specification.

[Total Marks: 50]